Statistics 250, Section 003, Final (300 Points)

May 12, 1999, Dr. Jürgen Symanzik

Your Name: ____________________

Question 1: Short Answers (**76 Points**)

1. For $x_1 = 1, x_2 = 5, x_3 = -20, x_4 = 0, x_5 = 10, x_6 = 4, x_7 = -4, x_8 = 4$, and $n = 8$, determine the following summary statistics: (**20 Points**)

   mean:

   median:

   range:

   sample variance:

   sample standard deviation:

2. For the numbers given in (1) above, determine the following sums: (**12 Points**)

   $$\sum_{i=1}^{(n/2)-1} x_i \cdot x_{n-i} =$$

   $$\sum_{i=n-2}^{n} i^2 \cdot x_{(i)} =$$

   $$\sum_{i=1}^{n-4} (i!) \cdot x_1 =$$

3. Let $z_i = \frac{x_i - \overline{x}}{s}$, $i = 1, \ldots, 8$, be the z–scores for the 8 values given in (1) above. You do not even have to calculate the individual values for each z_i to indicate which of the following statements is correct: (**6 Points**)

   (a) The correlation coefficient r between x and z must be negative since I subtract $\overline{x}$ from each x_i.

   (b) The correlation coefficient r between x and z is somewhere between 0.7 and 0.9.

   (c) The correlation coefficient r between x and z is exactly 1.

   (d) The correlation coefficient r between x and z is exactly 0.

4. Similar to the median (when compared to the mean), there exists a more robust measure of variability (when compared to the variance) that is not affected too easily by unusual large or small values. This value is called the *median absolute deviation* (MAD) and it is defined as **the median of the absolute deviations from the median**, in symbols

$$\text{MAD} = median\{\ |x_i - \tilde{x}|, \quad i = 1, \ldots, n\}.$$

   Calculate the MAD for the numbers given in (1) above. (**6 Points**)

5. Determine the slope and the y–intercept of the lines whose equations are given as: (**6 Points**)

   (a) $8x - 4y = 20$

   (b) $5y + 7x = 14$

6. The British lottery has a game called "7 out of 38". You make a selection of 7 numbers between 1 and 38 and you win the big prize if exactly these numbers are drawn in the weekly drawing. How many different combinations are possible to select 7 out of 38 numbers? Calculate this value! (**6 Points**)

7. What is wrong with this graphic from “Time” (April 9, 1979, p. 57), reprinted in Edward R. Tufte’s book “The Visual Display of Quantitative Information”. Provide a better graphical representation of the same data. (**8 Points**)

8. Let Z be a standard Normal variable, i.e., $Z \sim N(0,1)$, and X be a Normal variable with mean $\mu = 2$ and variance $\sigma^2 = 9$, i.e., $X \sim N(2, 3^2)$. Determine the following: (**12 Points**)

   (a) $P(Z < 1.85) =$

   (b) $P(-2.0 < Z < 1.0) =$

   (c) $P(X < 1.85) =$

   (d) $P(-2.0 < X < 1.0) =$

   (e) Find a number **#** so that $P(Z < \#) = 0.20$

   (f) Find a number **#** so that $P(X < \#) = 0.20$

Question 2: Probability Distributions (**53 Points**)

For homework assignment #8, Amanda reported the following point distribution for those 50 Stat 250 students that turned in this particular homework assignment:

# students	points
1	9
1	10
0	11
1	12
0	13
2	14
2	15
1	16
5	17
2	18
10	19
25	20

We introduce a random variable Y that represents the points for homework assignment #8.

1. Indicate the probability distribution $p(y)$ for Y in tabular form. (**10 Points**)

2. Draw a spike graph of $p(y)$. (**5 Points**)

3. Add the values for $F(y)$, i.e., the cumulative probability function for Y, to your table in (1.) above and draw a graph of $F(y)$. (**15 Points**)

4. What is $P(Y \leq 18)$? (**3 Points**)

5. Calculate $\mu = E(Y)$, i.e., the mean (expected value) of Y. (**10 Points**)

6. Calculate $\sigma^2 = Var(Y)$, i.e., the variance of Y. (**10 Points**)

Question 3: Scatterplot Matrix & Linked Brushing (**40 Points**)

The questions on the next page are based on the scatterplot matrix presented in William Cleveland's book "Visualizing Data". This scatterplot matrix has been reprinted below. It displays trivariate data that represents measurements of "Abrasion Loss", "Hardness", and "Tensile Strength" for 30 rubber specimens. "Abrasion Loss" is the dependent variable and "Hardness" and "Tensile Strength" are the independent variables. The goal of this study was to determine conditions that minimize the "Abrasion Loss".

1. Label the (individual) scatterplot that shows the "Tensile Strength" on the horizontal (x–)axis and the "Abrasion Loss" on the vertical (y–)axis with the letter "A". (**5 Points**)

2. What is the (approximate) range R of "Abraison Loss"? (**5 Points**)

3. Which of these statements is correct/incorrect?

   (a) The brush is located in the scatterplot that shows the "Tensile Strength" on the vertical (y–)axis and the "Hardness" on the horizontal (x–)axis. (**5 Points**)

   (b) Since a straight line can be fitted reasonably well to the data in any of the scatterplots that shows "Abrasion Loss" and "Hardness", the correlation coefficient r between these two variables must be positive, i.e., $r > 0$. (**5 Points**)

   (c) The low values (i.e., values in the approximate range 120–165) of "Hardness", have been brushed. (**5 Points**)

   (d) When "Hardness" is fixed at any given level, lower values of "Tensile Strength" result in a much higher value for "Abrasion Loss" than higher values of "Tensile Strength". (**5 Points**)

4. Decide on **ONE** of the following options. The best way to minimize "Abrasion Loss" is a combination of (**10 Points**)

   (a) low "Hardness" and low "Tensile Strength"

   (b) low "Hardness" and high "Tensile Strength"

   (c) high "Hardness" and low "Tensile Strength"

   (d) high "Hardness" and high "Tensile Strength"

Question 4: Binomial Probability Distributions (**55 Points**)

In Quiz 3, only 5 out of 50 Stat 250 students answered the statistic/parameter question correctly. Let's assume that in the upcoming Fall semester all new Stat 250 students have to answer this same question in one of their quizzes. Based on previous years, the upper enrollment limit of 200 students has been obtained each year so we can safely assume that there will be 200 students again this fall that will attend this class.

Let X be the random variable that describes the number of students that will correctly answer this question in the fall semester. It is safe to asssume that students' ability to answer a particular question does not change over time.

1. Indicate the probability distribution of X. Complete the following formula that relates to the probability distribution of X: (**5 Points**)

$$X \sim$$

2. Calculate the mean and the variance of X. (**20 Points**)

3. What is the **exact** probability that 20 students will answer this question correctly? Be as efficient in your calculations as possible. (**10 Points**)

4. What is the probability that 30 through 50 students will answer this question correctly? You may use an **approximation** if **appropriate**. (**20 Points**)

Question 5: Linear Regression & Correlation (**76 Points**)

The average prices (in dollars) per ounce of gold and silver for the years 1986 through 1994 are given below. (Source: U.S. Bureau of Mines.)

Year	Gold	Silver
1986	368	5.47
1987	478	7.01
1988	438	6.53
1989	383	5.50
1990	385	4.82
1991	363	4.04
1992	345	3.94
1993	361	4.30
1994	389	5.30

1. Draw a scatterplot where x is the silver price and y is the gold price. Does the appearance of the scatterplot suggest a linear relationship between x and y? (**7 Points**)

2. Fit a least squares (linear regression) line to the data. Make clear what your variables stand for. It might help to know that:

$$\sum_{i=1}^{n} x_i = 46.91, \sum_{i=1}^{n} x_i^2 = 253.6095, \sum_{i=1}^{n} y_i = 3,510, \sum_{i=1}^{n} y_i^2 = 1,383,102, \sum_{i=1}^{n} x_i y_i = 18,625.9$$

(**30 Points**)

3. Calculate Pearson's correlation coefficient r between x and y. How can we interpret this value for our given data set? (**10 Points**)

4. Based on your calculations in (2), what is the predicted price of an ounce of gold when the price of an ounce of silver is \$5.00? Do you think it is safe to predict that an ounce of gold costs about \$237 if the price of an ounce of silver drops to \$1.00? Explain. (**15 Points**)

5. Indicate which residual plots should be plotted for this data set. You should describe **at least 2** different residual plots that might be useful here. You **do not** have to draw any such plot to get full credit. (**14 Points**)